

wait=-6746864, due=1695531540, due2=0 Question number 1.

Your answer was B. Correct. 

Give the differential equation that has $y = C_1 + C_2 x^3 + x$ as its general solution.

A $x y'' + 2y' = 2$

B $x y'' - 2y' = -2$

C $x y'' - 2y' = 4$

D $x y' - 2y = -2$

E $x y'' + 4y' = -2$

F None of the above.

Question number 2.

Your answer was A. Correct. 

Find the general solution of $x y' = 3x^4 e^{2x} + 3y$.

A $y = \frac{3}{2}x^3 e^{2x} + Cx^3$

B $y = \frac{3}{2}x^{-3} e^{2x} + Cx^{-3}$

C $y = \frac{1}{3}x^3 e^{2x} + Cx^3$

D $y = \frac{3}{2}x^3 e^{-2x} + Cx^3$

E $y = 3x^3 e^{2x} + Cx^3$

F None of the above.

Question number 3.

Your answer was A. Correct.



Find the general solution of $(xy^3 - 2x)y' = x^2y + 2y$.

A $\frac{1}{3}y^3 - 2\ln|y| = \frac{1}{2}x^2 + 2\ln|x| + C$

B $y^3 - 2\ln|y| = x^2 + 2\ln|x| + C$

C $\frac{1}{3}y^3 - 2y = \frac{1}{2}x^2 + 2x + C$

D $\frac{1}{3}y^3 + \frac{2}{y^{-2}} = \frac{1}{2}x^2 - \frac{2}{x^{-2}} + C$

E $\frac{1}{3}y^4 - 2\ln|y| = \frac{1}{2}x^3 + 2\ln|x| + C$

F None of the above.

Question number 4.

Your answer was D. Wrong. 

The function $y = 2e^{3x} + 3xe^{3x}$ is a solution of second order linear homogeneous differential equation with constant coefficients. The differential equation is:

A $y'' + 6y' + 13y = 0$

B $y'' - 6y' + 9y = 0$

C $y'' - 5y' + 6y = 0$

D $y'' + 6y' + 9y = 0$

E $y'' + 3y' + 5y = 0$

F None of the above.

Question number 5.

Your answer was B. Correct. 

A fundamental set of solutions of $y'' + 7y' + 10y = 0$ is:

A $\{e^{-2x}, e^{5x}\}$

B $\{e^{-2x}, e^{-5x}\}$

C $\{e^{-2x}, xe^{-2x}\}$

D $\{e^{2x}, e^{-5x}\}$

E $\{e^{2x}, e^{5x}\}$

F None of the above.

Question number 6.

Your answer was E. Wrong. 

Give the general solution of $y'' - 6y' + 13y = 0$.

A $y = C_1 e^{3x} + C_2 x e^{3x}$

B $y = C_1 e^{3x} \cos(2x) + C_2 e^{3x} \sin(2x)$

C $y = C_1 e^{-3x} \cos(2x) + C_2 e^{-3x} \sin(2x)$

D $y = C_1 e^{3x} + C_2 e^{2x}$

E $y = C_1 e^{2x} \cos(3x) + C_2 e^{2x} \sin(3x)$

F None of the above.

Question number 7.

Your answer was A. Wrong. 

The differential equation for the family of curves $y^3 = Cx^4 - 2$ is

$$y' = \frac{4y^3 + 8}{3xy^2}.$$

Find the family of orthogonal trajectories.

A $3x^2 + 4y^2 + \frac{16}{y} = C$

B $3x^2 + 4y^2 + \frac{32}{y} = C$

C $3x^2 + 4y^2 + 16 \ln(y) = C$

D $3x^2 + 4y^2 - 16 \ln(y) = C$

E $3x^2 + 4y^2 - \frac{16}{y} = C$

F None of the above.

Question number 8.

Your answer was A. Correct.



This is a written question, worth 10 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.

Problem Code: 872

(a) In each of (1) and (2), determine whether the given equation is linear, separable, Bernoulli, homogeneous, or none of these.

(1) $x^2 y' = 2x^2 + xy + 2y^2$

(2) $xy' = 2x^2y + 3x^4 \sin(2y^2 + 1)$

(b) Find the general solution of $y' + \frac{4}{x}y = \frac{4e^{2x}}{x^2} \sqrt{y}$.

A I have placed my work and my answer on my answer sheet.

B I want to have points deducted from my test for not working this problem.

Question number 9.

Your answer was B. Wrong.



This is a written question, worth 10 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.

Problem Code: 964

Two years ago a laboratory was given a 50 gram sample of a radioactive substance. Now there are 40 grams of the substance.

- (a) Find the size of the sample at any time t .
- (b) What will the size of the sample be 2 years from now?
- (c) What is the half-life of the substance?

- A I have placed my work and my answer on my answer sheet.
- B I want to have points deducted from my test for not working this problem.

Question number 10.

Your answer was A. Correct. 

This is a written question, worth 10 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.

Problem Code: 1065

Given the initial-value problem

$$y'' + \frac{3}{x}y' - \frac{8}{x^2}y = 0, \quad y(1) = 4, \quad y'(1) = 2.$$

- (a) The differential equation has solutions of the form $y = x^r$. Find two independent solutions y_1, y_2 of the equation. Use the Wronskian to justify your answer.
- (b) Give the general solution of the differential equation.
- (c) Find the solution that satisfies the initial conditions.

- A I have placed my work and my answer on my answer sheet.

B I want to have points deducted from my test for not working this problem.

Question number 11.

Your answer was A. Correct. 

This is a BONUS written question, worth 2 points. DO NOT place the problem code on the answer sheet. A proctor will fill this out after exam submission. Show all steps (work) on your answer sheet for full credit.

Problem Code: 1124

A disease is infecting a colony of 1000 penguins living on a remote island. Let $P(t)$ be the number of sick penguins t days after the outbreak. Suppose that 25 penguins had the disease initially, and suppose that the disease is spreading at a rate proportional to the product of the time elapsed and the number of penguins which do not have the disease.

Give the mathematical model (initial-value problem) for P .

A I have placed my work and my answer on my answer sheet.

B I want to have points deducted from my test for not working this problem.
